LMER Model Predictions

Erik Larsen

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Overview

This vignette shows the linear mixed model used to measure my performance for each golf round and use that information to predict my performance for the next golf round

It does this by incorporating my skill level and where I've played in the past

Skill level is officially determined by the **USGA** as a `Handicap Index`: how many shots, on average, a player takes to complete a round at a given course with a given difficulty relative to that course's average

- i.e. a player with a **10.0** `Handicap Index` is expected, on average, to take **82 shots** to complete a round at a course where the average number of strokes has been determined to be **72**

`Gross Score` is a total number of shots hit by a given player for any individual hole

- this is extrapolated across each of 18 holes, so `Gross Score` can mean a total of shots for a hole *or* a total of shots for a round
- the average `Gross Score` over a player's best **8 rounds** *from their last 20* is used to determine their `Handicap Index`

Thus, this vignette shows the model used to predict the opposite:

- given my `Handicap Index` and `Gross Score`s at previous courses of varying difficulty in the past,
  - **what will be my next** `Gross Score`?

Note that I use the terms, `strokes`, and, `shots`, interchangeably, though putts are not connoted as `shots`

Code Environment Details

```r
library(golf)
library(tidyverse)
library(lme4)
library(mgcv)
library(brms)
library(DBI)
library(RSQLite)
library(emayili)
```

Attach Packages

Code for Summarizing Metrics

Gather and Format Scores Gather and format from the database

```
## connect to the db
con <- golf::get_db_connection()

scores <- DBI::dbGetQuery(conn = con, statement = paste0(
  "SELECT DISTINCT sub.* FROM
  (SELECT DISTINCT r.*, c.par, c.course_rating FROM rounds r
  LEFT JOIN courses c
  ON c.tees = r.tees
  AND c.course_name = r.course_name
  AND c.hole = r.hole
  AND c.hole_handicap = r.hole_handicap) AS sub
  INNER JOIN players p
  ON sub.GHIN = p.GHIN
  AND sub.handicap_index = p.handicap_index
  AND sub.date = p.date;"
)) |>
  dplyr::mutate(date = lubridate::as_date(date),
              hole = stringr::str_extract(hole, pattern = '[0-9]{1,}') |>
                as.numeric()) |>
  dplyr::relocate(par, .after = hole) |>
  dplyr::relocate(course_rating, .after = tees) |>
  dplyr::group_by(date) |>
  dplyr::arrange(desc(date), hole) |>
  dplyr::ungroup()
```

Code for Computing Advanced Metrics

Compute Metrics Compute more nuanced metrics

Advanced Metrics Snapshot

```
head(scores_sum |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 26
## # Groups:   GHIN, date, date_course, course_rating [6]
##      GHIN date       date_course course_rating `Handicap Index`  FIRs `Iron FIRs` `Iron FIR%` `Drive
##     <int> <date>     <chr>               <dbl>            <dbl> <int>       <dbl>       <dbl>
## 1  1.05e7 2026-04-05 "2026-04-0~          71.7               11     6           0           0
## 2  1.05e7 2026-03-29 "2026-03-2~          70.3               10     5           2          66.7
## 3  1.05e7 2026-03-08 "2026-03-0~          71.7               10     2           0           0
## 4  1.05e7 2026-02-22 "2026-02-2~          68                 10.1   7           1          33.3
## 5  1.05e7 2026-02-08 "2026-02-0~          70                 10.2   2           0           0
## 6  1.05e7 2026-01-25 "2026-01-2~          70                 10.2   4           0          NaN
## # i 4 more variables: `doubles+` <int>, penalties <int>, `Gross Score` <dbl>, `Net Score` <dbl>
```

Group Metrics

Separate the metrics:

Scoring Metrics Round scores and `Handicap Index`

```
scoring_metrics <- scores_sum |>
  dplyr::select(`Handicap Index`, `Gross Score`, `Net Score`)
head(scoring_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 7
## # Groups:   GHIN, date, date_course, course_rating [6]
##       GHIN date       date_course                            course_rating `Handicap Index` `Gross Score
##      <int> <date>     <chr>                                          <dbl>            <dbl>         <dbl
## 1 10526424 2026-04-05 "2026-04-05\nRandolph North\n11"               71.7             11              88
## 2 10526424 2026-03-29 "2026-03-29\nDell Urich\n10"                   70.3             10              82
## 3 10526424 2026-03-08 "2026-03-08\nRandolph North\n10"               71.7             10              84
## 4 10526424 2026-02-22 "2026-02-22\nDell Urich\n10.1"                 68               10.1            92
## 5 10526424 2026-02-08 "2026-02-08\nRandolph North\n10.2"             70               10.2            83
## 6 10526424 2026-01-25 "2026-01-25\nRandolph North\n10.2"             70               10.2            85
```

Stroke Metrics Above/below par

```
stroke_metrics <- scores_sum |>
  dplyr::select(`doubles+`, bogies, pars, birdies)
head(stroke_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 8
## # Groups:   GHIN, date, date_course, course_rating [6]
##       GHIN date       date_course                            course_rating `doubles+` bogies  pars birdi
##      <int> <date>     <chr>                                          <dbl>      <int>  <int> <int>   <in
## 1 10526424 2026-04-05 "2026-04-05\nRandolph North\n11"               71.7          3      9     6
## 2 10526424 2026-03-29 "2026-03-29\nDell Urich\n10"                   70.3          4      5     7
## 3 10526424 2026-03-08 "2026-03-08\nRandolph North\n10"               71.7          3      6     9
## 4 10526424 2026-02-22 "2026-02-22\nDell Urich\n10.1"                 68            6      7     5
## 5 10526424 2026-02-08 "2026-02-08\nRandolph North\n10.2"             70            2      8     7
## 6 10526424 2026-01-25 "2026-01-25\nRandolph North\n10.2"             70            2      9     7
```

Around-the-Green Metrics Chips, putts, etc.

```
atg_metrics <- scores_sum |>
  dplyr::select(chips, `chips+putts`, `UpDown%`, putts, `Avg GIR putts`)
head(atg_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 9
## # Groups:   GHIN, date, date_course, course_rating [6]
##       GHIN date       date_course                            course_rating chips `chips+putts` `UpDown%`
```

```
##      <int> <date>     <chr>                                          <dbl> <dbl>          <dbl>      <dbl>
## 1 10526424 2026-04-05 "2026-04-05\nRandolph North\n11"             71.7    21             52         20
## 2 10526424 2026-03-29 "2026-03-29\nDell Urich\n10"                 70.3    15             46         20
## 3 10526424 2026-03-08 "2026-03-08\nRandolph North\n10"             71.7    12             45         30
## 4 10526424 2026-02-22 "2026-02-22\nDell Urich\n10.1"               68      18             52         16.7
## 5 10526424 2026-02-08 "2026-02-08\nRandolph North\n10.2"           70      11             46         22.2
## 6 10526424 2026-01-25 "2026-01-25\nRandolph North\n10.2"           70      15             51         18.2
```

Ball Striking Approach and tee accuracy

```
ball_striking_metrics <- scores_sum |>
  dplyr::select(GIRs, `GIR%`, `Par 3 GIRs`,
                FIRs, `FIR%`, `Iron FIRs`, `Iron FIR%`,
                `Driver FIRs`, `Driver FIR%`)
head(ball_striking_metrics |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 13
## # Groups:   GHIN, date, date_course, course_rating [6]
##       GHIN date       date_course                        course_rating  GIRs `GIR%` `Par 3 GIRs`  FI
##      <int> <date>     <chr>                                      <dbl> <int>  <dbl>        <dbl> <in
## 1 10526424 2026-04-05 "2026-04-05\nRandolph North\n11"            71.7     3   16.7            1
## 2 10526424 2026-03-29 "2026-03-29\nDell Urich\n10"                70.3     7   38.9            1
## 3 10526424 2026-03-08 "2026-03-08\nRandolph North\n10"            71.7     7   38.9            2
## 4 10526424 2026-02-22 "2026-02-22\nDell Urich\n10.1"              68       3   16.7            1
## 5 10526424 2026-02-08 "2026-02-08\nRandolph North\n10.2"          70       7   38.9            3
## 6 10526424 2026-01-25 "2026-01-25\nRandolph North\n10.2"          70       6   33.3            2
```

Shot Quality Yardage and accuracy on tracked shots

```
## connect to the db
con <- golf::get_db_connection()

stroke_quality <- DBI::dbGetQuery(conn = con,
                                  statement = paste0("SELECT DISTINCT * FROM club_metrics;")) |>
  dplyr::mutate(date = lubridate::as_date(date),
                hole = gsub(hole, pattern = 'hole_', replacement = ''),
                hole = as.integer(hole)) |>
  dplyr::mutate(dplyr::across(dplyr::contains("yds"), ~as.integer(.x))) |>
  dplyr::rename(course = course_name) |>
  dplyr::group_by(date, hole, stroke) |>
  dplyr::arrange(date, hole, stroke)
head(stroke_quality |>
       dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 14
## # Groups:   date, hole, stroke [6]
##   course         date       tees   hole   par gross stroke lie     club  yds_to_target yds_traveled
##   <chr>          <date>     <chr> <int> <int> <int>  <int> <chr>   <chr>         <int>        <int>
## 1 Randolph North 2026-04-05 blue      1     4     4      1 tee     4               220          219
## 2 Randolph North 2026-04-05 blue      1     4     4      2 rough   PW              140          131
## 3 Randolph North 2026-04-05 blue      1     4     4      3 fairway PW               10            8
```

```
## 4 Randolph North 2026-04-05 blue      2     4     6      1 tee      D                 270            308 
## 5 Randolph North 2026-04-05 blue      2     4     6      2 rough    4                 115             82 
## 6 Randolph North 2026-04-05 blue      2     4     6      3 sand     SW                 30             16 
```

LMER Model

Fit a LMER Model Fit a lmer model to capture repeated measurements of `Gross Score` predicted by `Handicap Index`, `course_rating`, and time (`days`).

- Every course has a rating; the `Handicap Index` calculation factors in these ratings
- Center the `course_rating` and `Handicap Index` variables at their mean to make interpreting intercept and slope estimates more meaningful
- Include random intercepts and random slopes of `course` and `course_rating`, given a `Handicap Index`

```r
# Fit a model to the data

gross_lmer <- lme4::lmer(

  data = scores_sum |>
    dplyr::ungroup() |>
    dplyr::mutate(

      course_rating = mean(course_rating) - course_rating,

      course = gsub(date_course,
                    pattern = '[0-9]|\\-|\\\n|\\.',
                    replacement = ''), # extract the course names

      `Handicap Index` = mean(`Handicap Index`) - `Handicap Index`,
      days = as.numeric(as.Date(date) - min(as.Date(date)) + 1,
                        units = 'days')
      ) |> # create a 'days' metric starting at the first day joining the club

    dplyr::relocate(days, .after = date),

  formula =
    `Gross Score` ~
    `Handicap Index` +
    course_rating +
    course +
    days +
    (1 + `Handicap Index`|course) + # random intercepts and random slopes for Gross Score at a course g
    (1 + `Handicap Index`|course_rating) # random intercepts and random slopes for Gross Score at a cou
           )
```

LMER Model Summary

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: `Gross Score` ~ `Handicap Index` + course_rating + course + days +      (1 + `Handicap Inde
##    Data: dplyr::relocate(dplyr::mutate(dplyr::ungroup(scores_sum), course_rating = mean(course_rating
##     replacement = ""), `Handicap Index` = mean(`Handicap Index`) -      `Handicap Index`, days = as.
```

```
##
## REML criterion at convergence: 143.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.3923 -0.6164 -0.2016  0.5644  1.8382
##
## Random effects:
##  Groups        Name             Variance Std.Dev. Corr
##  course_rating (Intercept)      21.4784  4.6345
##                `Handicap Index`  0.9569  0.9782   -1.00
##  course        (Intercept)      17.7971  4.2187
##                `Handicap Index` 12.0015  3.4643   0.09
##  Residual                        9.0663  3.0110
## Number of obs: 30, groups:  course_rating, 7; course, 5
##
## Fixed effects:
##                      Estimate Std. Error t value
## (Intercept)          93.654681   9.349804  10.017
## `Handicap Index`      2.535617   2.278252   1.113
## course_rating         0.081533   1.462834   0.056
## courseDell Urich     -4.945540  10.273306  -0.481
## courseRandolph North -0.007814  11.700073  -0.001
## courseSewailo         2.510066  14.641846   0.171
## courseSilverbell     -3.787574  10.502635  -0.361
## days                 -0.034983   0.010627  -3.292
##
## Correlation of Fixed Effects:
##             (Intr) `HInd` crs_rt crsDlU crsRnN crsSwl crsSlv
## `HndcpIndx`  0.181
## course_rtng -0.606 -0.106
## corsDllUrch -0.826 -0.162  0.454
## crsRndlphNr -0.866 -0.143  0.635  0.748
## courseSewal -0.558  0.239  0.331  0.526  0.494
## corsSlvrbll -0.834 -0.171  0.468  0.773  0.739  0.578
## days        -0.342 -0.217  0.350  0.078  0.146  0.095  0.168
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
golf::export_lm_round_predictions(
  model = gross_lmer,
  scores_sum = scores_sum |>
    dplyr::mutate(date = as.character(date))
)
```

Export Model Results to db

Model Interpretations

Big Picture: Handicap Index as a Fixed Effect Predicting Gross Score The model's aggregate average `Gross Score` (**`(Intercept)` Estimate of Fixed effects**) is **93.65**. This is model's estimated average first `Gross Score`.

My average `Gross Score`, however, is **85.2**.

For every additional `Handicap Index` point larger than the average `Handicap Index`, my `Gross Score` increases by **2.54** strokes.

- This makes sense because `Gross Score` is directly used to determine `Handicap Index` and is positively correlated (high `Gross Score` = high `Handicap Index`)
  - In other words, a player with great skill will have a low `Handicap Index` (i.e. **0**), meaning they average `par` (the course average) for an entire round, while a worse player (who averages above `par`, the course average) will have a higher `Handicap Index`
    * We know this to be true
    * `Handicap Index` effectively corrects for skill-level to determine who performed better that day
- While this makes sense, I wonder whether `Handicap Index` should have a larger `Fixed effect Estimate`, which essentially asks, "how much does this variable predict the target (`Gross Score`, here)?"
  - The effect is significant (**`t value` = 1.11**; significance : abs(**t value**) > 1)
  - Again, `Handicap Index` is a metric *directly derived from* `Gross Score`, thus, I'm unsure how many strokes (`Gross Score`) index points should be worth! **1? More?** Does it vary by skill, or is it uniform?

A Deeper Dig: Course, Course Rating, and Time For every additional `course_rating` point (aka, a stroke) greater than the average `course_rating` (~69-70 strokes in this dataset), `Gross Score` increases by **0.08** strokes.

- This also makes sense: harder courses should yield higher `Gross Score`s
  - These courses vary in their difficulty, independent of player skill (`Handicap Index`), by **21.48** strokes, on average, even though `course_rating` is supposed to account for course difficulty across all courses. This is the **`Random effects Variance (Intercept)`**.
  - When compared to the `Residual Variance`, **9.07**, a `course_rating` variance of **21.48** is very high– I play differently according to `course_rating`

The `course` also has an effect on `Gross Score`: I play more consistently at some courses than others, as suggested by the large variance of **17.8** strokes

For every additional `day` in time, my `Gross Score` drops by **-0.03** strokes.

- While this seems tiny, extrapolating days to months or weeks, this becomes very evident (**-0.9** strokes per month; **-10.95** strokes per year)
- Linear extrapolation in this sense is misleading: there will be a limit to lowering `Gross Score` and there will also be variation in the process
- But this effect is strongly significant (**t value = -3.29**) and appears to be the primary driver of the trend

Predict the Next Round

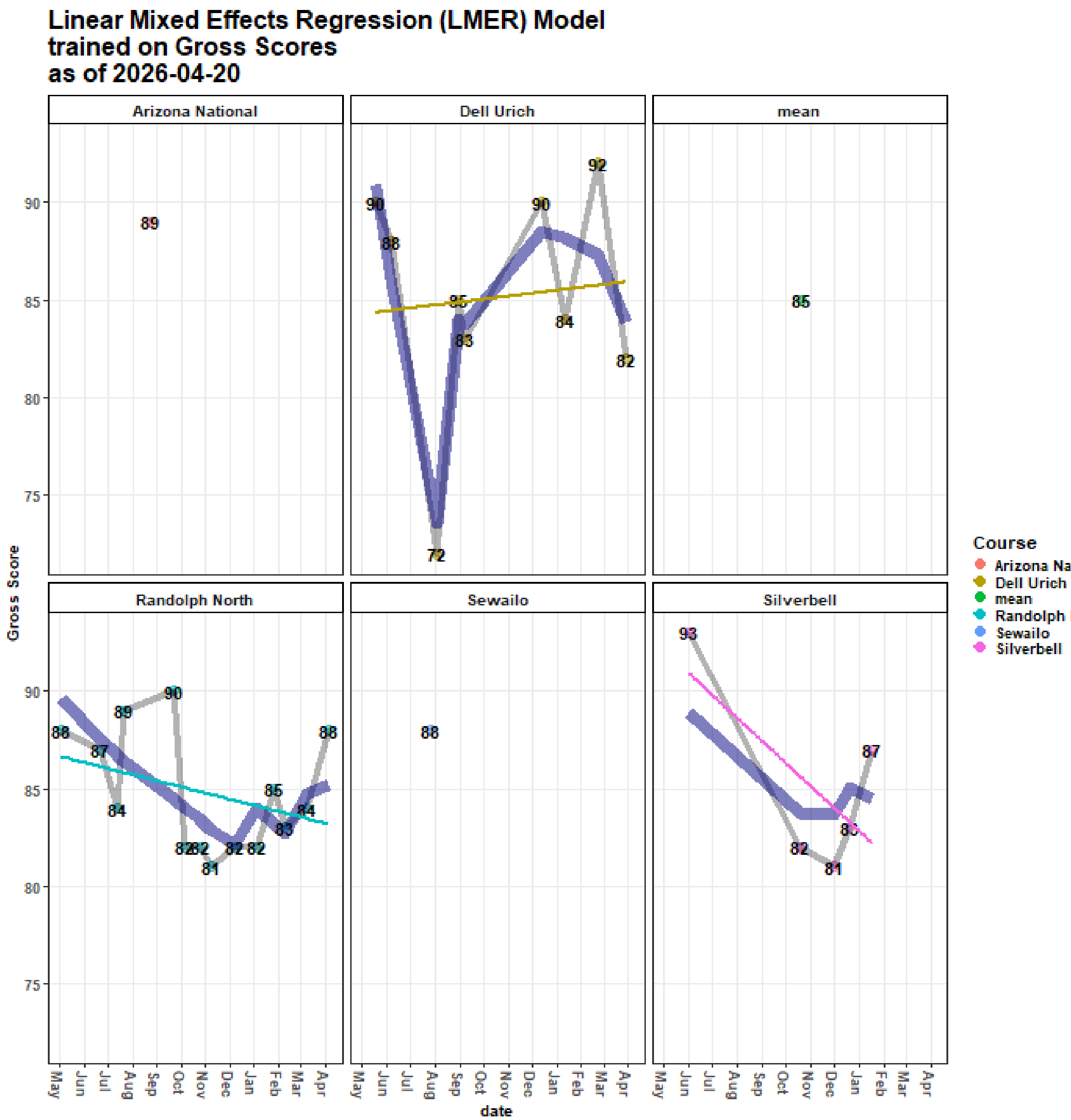
Predict the next round's `Gross Score` according to the model

```r
## show the model-predicted gross score for the upcoming round, rounded to the nearest stroke
stats::predict(object = gross_lmer, newdata = new_df, allow.new.levels = T) |>
  as.numeric() %>%
  round(., 0)
```

Show Prediction

```
## [1] 81
```

Plot the Model



Model by Course

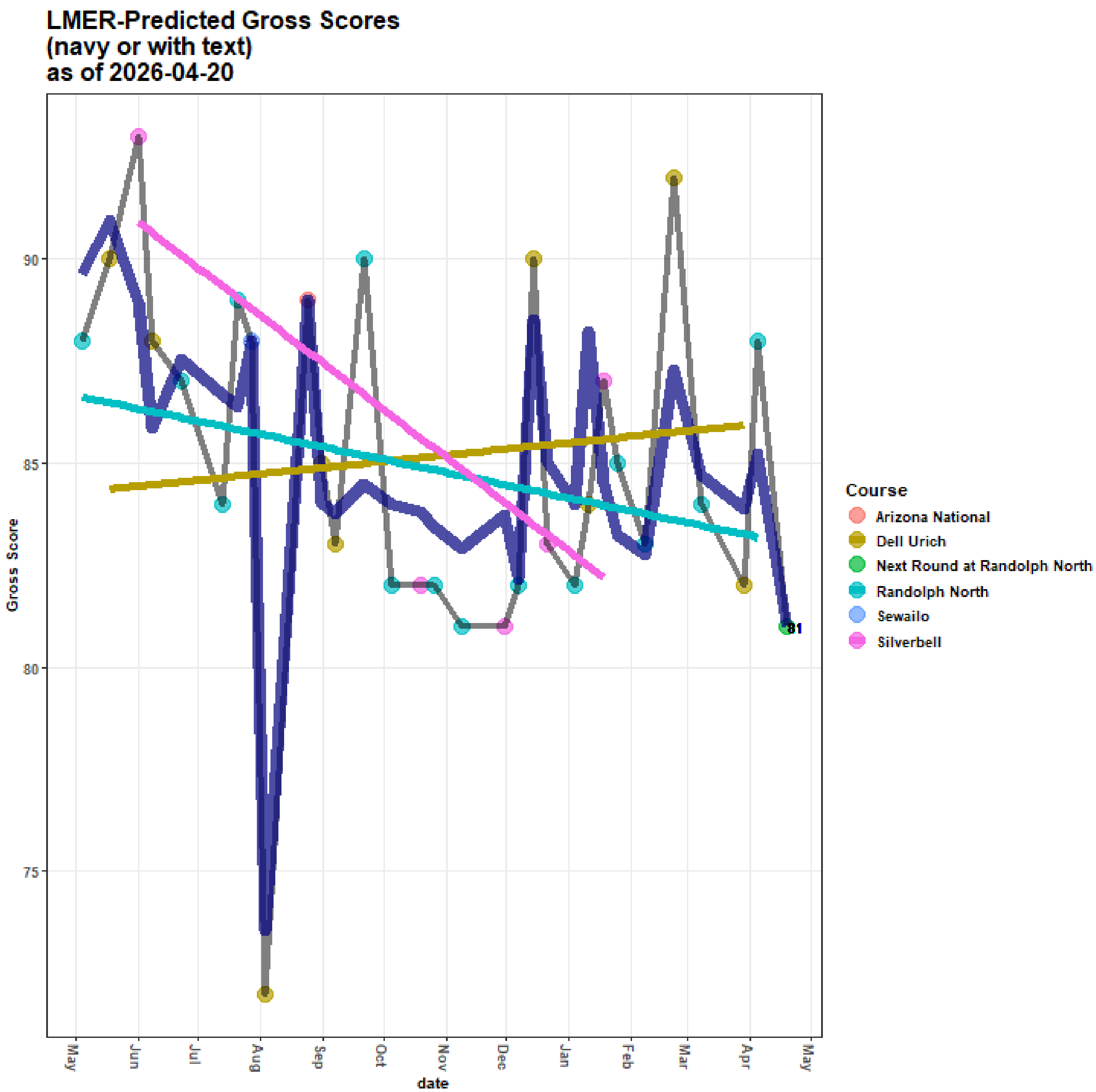
The model is a random intercept, random slope linear mixed-effects regression (LMER) model.

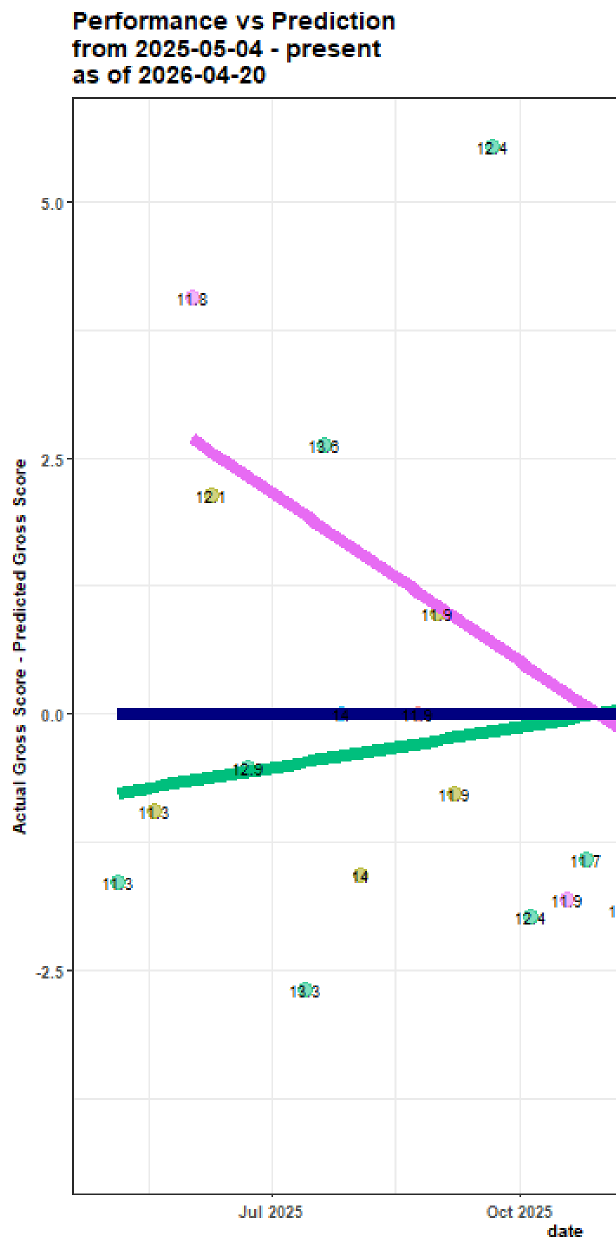
In this case, that means `Gross Score` varies for each course at a given `Handicap Index` in its deviation from the overall mean `Gross Score` (navy blue line) over time: `Silverbell`, `Randolph North`, and `Dell Urich` have their own average `Gross Scores` (intercepts) and slopes (change in `Gross Score` over time)– notice how the line for each course has a different slope, starting at a different y-intercept

- The `blue` line is the model's overall fit of the `Gross Score`, accounting for `course`, `course_rating`, `Handicap Index`, and `days` (time)
- `Silverbell`'s line represents the relationship between `Gross Score`, `course_rating`, `Handicap Index`, and `days` (date/time) at `Silverbell`
- `Randolph`'s line represents the relationship between `Gross Score`, `course_rating`, `Handicap Index`, and `days` (date/time) at `Randolph North`
- `Dell Urich`'s line represents the relationship between `Gross Score`, `course_rating`, `Handicap Index`, and `days` (date/time) at `Dell Urich`

Model Predictions These predictions are **not** historical– they are current:

- i.e. the model did *not* predict anything close to the **72** in August 2025


LMER-Predicted Gross Scores
(navy or with text)
as of 2026-04-20
Gross Score
90
85
80
75
date
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
81
Course
Arizona National
Dell Urich
Next Round at Randolph North
Randolph North
Sewailo
Silverbell

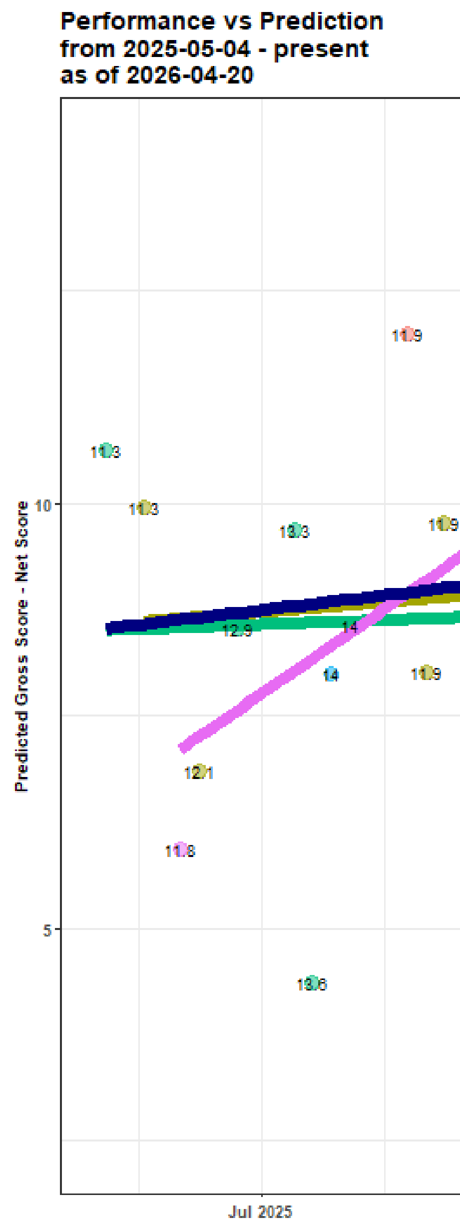


Actual Gross vs Predicted Gross (performance vs prediction)

This plot of residuals reveals the `Actual Gross Score` relative to the `Predicted Gross Score` over time, color-coded by `course`, and annotated by `Handicap Index` at the time of the round.

- the navy blue line represents the dividing line between over/under performing where:
  - scores **above** the line = I performed **worse** than the model's prediction
  - scores **below** the line = I performed **better** than the model's prediction
- `Randolph North`, `Silverbell`, and `Dell Urich` each have lines representing the trend of actual `Gross Scores` compared to predicted `Gross Scores` at each respective course
  - I more often score better/lower at `Randolph North` than the model predicts, though, on average, these are closest to the model's predictions
    - This might reveal that these rounds are contributing more weight to the model if they are not simply more reflective of the overall trend

    * This could also reveal that I score more consistently at this course than others
    * This could also mean that, given the downward trend, the course is easier than ratings suggest
  - At `Silverbell` and `Randolph North`, I have been outperforming the model and scoring better over time
  - The variability at `Dell Urich` is substantial, with one large outlier overperformance (~ **-15**), and one moderate outlier underperformance (~ **+6**) re-shaping the slope in the opposite direction: I have been getting worse at `Dell Urich` over time
    * Even with more subsequent sample, this could reveal that I struggle to shoot low `Gross Scores` at `Dell Urich`, despite its easier course rating, and any effect of time
    * Other latent variables may contribute to this variability, such as course/weather/event conditions



Actual Net vs Predicted Gross (skill-adjusted performance vs prediction)

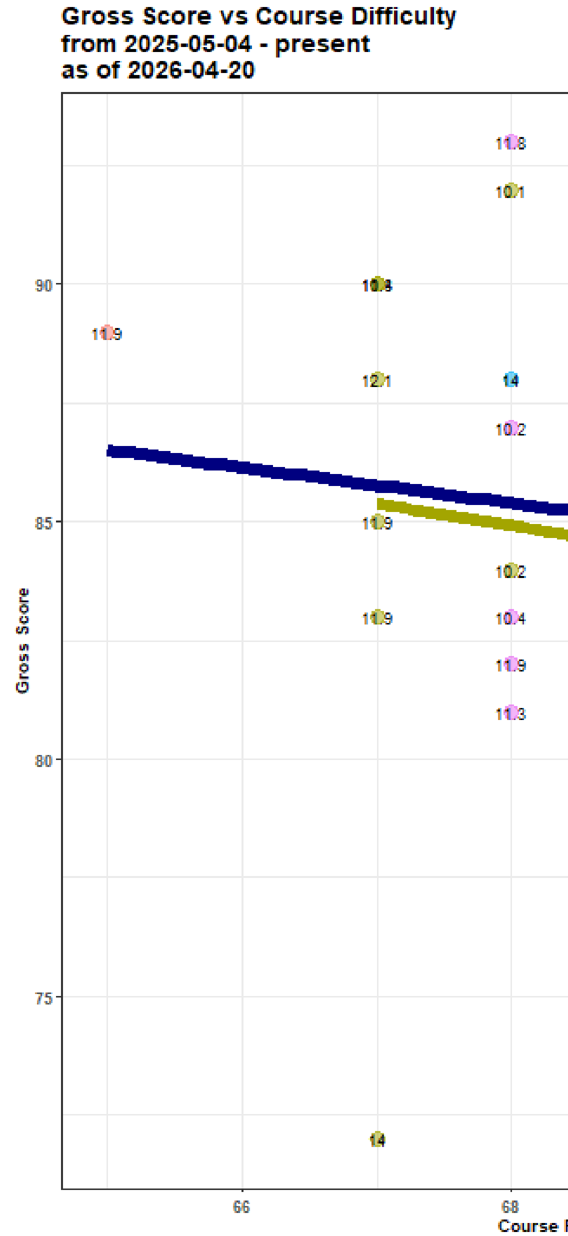
Relative to the previous plot, this plot of residuals:

- shifts the previous plot upward
- inverts it about the x-axis
- rotates it slightly about the origin

It shows the `Actual Net Score` relative to the `Predicted Gross Score` over time, color-coded by `course`, and annotated by `Handicap Index` at the time of the round.

`Net Score` is roughly `Gross Score - Handicap Index`.

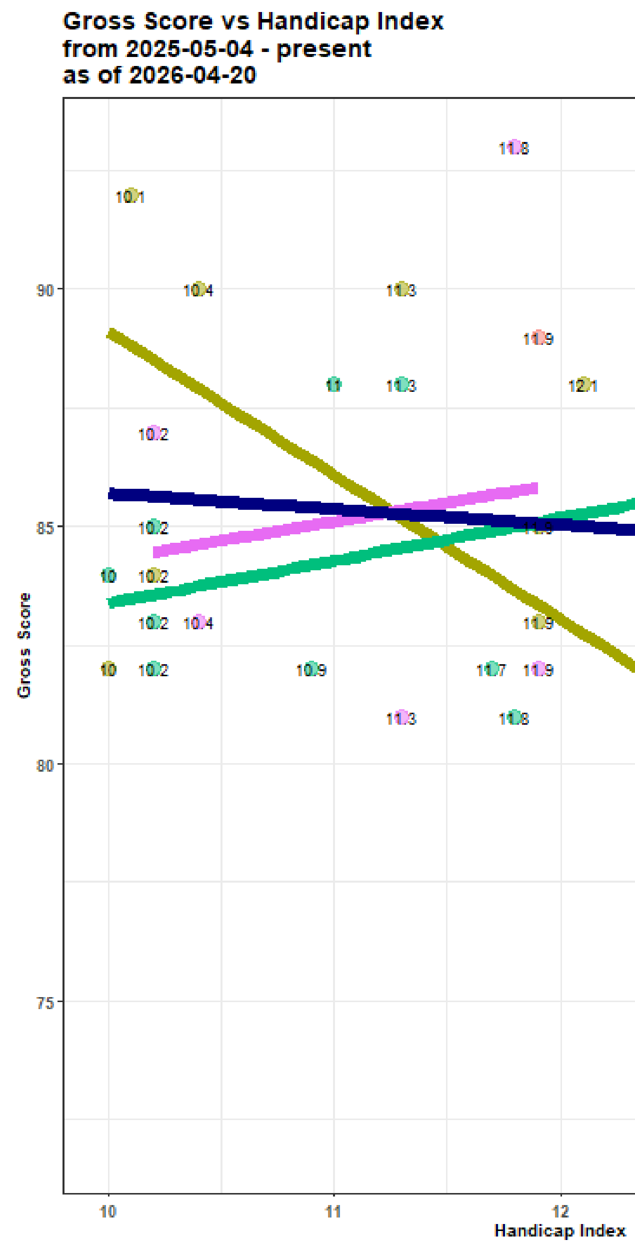
- the navy blue line represents the dividing line between over/under performing where:
  - scores **below** the line = I performed **worse** than the model's prediction
  - scores **above** the line = I performed **better** than the model's prediction
- `Randolph North`, `Silverbell`, and `Dell Urich` each have lines representing the trend of actual `Net Score`s compared to `Predicted Gross Score`s at each respective course
  - I more often score better/lower at `Randolph North` than the model predicts, particularly over time, given my handicap; however, on average, these are closest to the model's predictions than other courses
    - This reveals that these rounds could be giving more weight to the model
    - This could also reveal that I more consistently, if slightly, outscore the model at this course, even given my `Handicap Index`
    - This could also mean that the course is easier than ratings suggest
    - My instinct is that my `Handicap Index` was overestimated early on in this time series; it was high, and I frequently played `Randolph North` around then, and shot lower scores, directing the trend downward
  - The variability at `Dell Urich` is substantial, with one large outlier overperformance (~ **-15**), and one moderate outlier underperformance (~ **+6**) re-shaping the slope in the opposite direction (positive as opposed to a negative slope, like `Silverbell` and `Randolph North`)
    - Even with more subsequent sample, this could reveal that I struggle to shoot low `Gross Score`s at `Dell Urich`, despite its easier course rating
      - See the plot below for more insight
    - Other latent variables may contribute to this variability, such as course/weather/event conditions



Actual Gross vs Course Rating (performance vs course difficulty)

This definitely shows that I struggle at `Dell Urich`– independent of time, my `Gross Score`s at `Dell Urich` are roughly similar to other courses despite its easier rating– this would be even more evident without the substantial `Gross Score` **`72`** outlier.

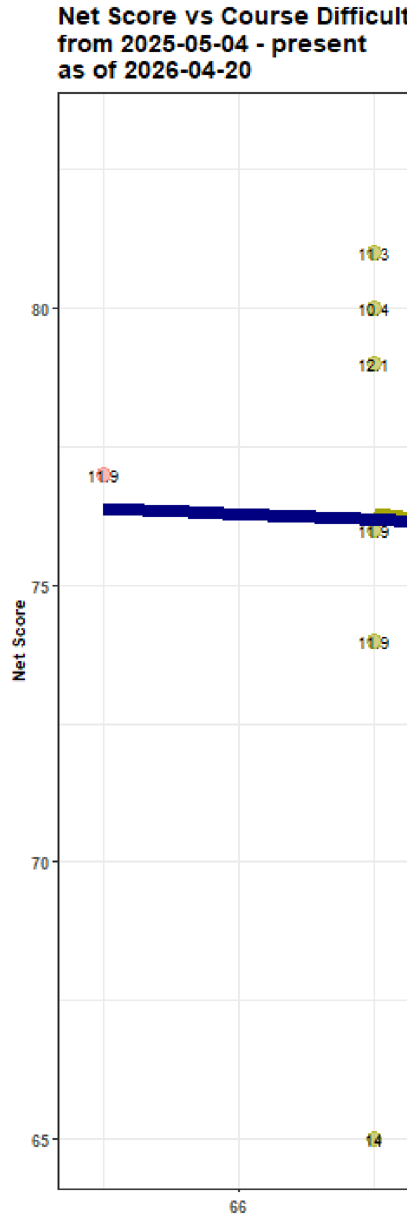
- Interestingly, I have scored better at longer/more difficult tees at multiple courses.
- Removing the effect of time/improved skill, and the wildly underrated `Arizona National` rating, this would otherwise capture the general trend and logic that **`higher course ratings correlate to higher Gross Scores`**



Actual Gross vs Handicap Index (performance vs skill-level)

This also supports the ideas that, independent of time and `Handicap Index`, I struggle at `Dell Urich` because of the high `Gross Score`s at low `Handicap Index`:

- Removing the outlier at a **`Handicap Index of 14`**, the `Dell Urich` trend still doesn't reverse, though the overall trend does– independent of time and one outlier/corrective round, I perform worse at a course with a lower `Handicap Index`.



Actual Net vs Course Rating (skill-adjusted performance vs course difficulty)